

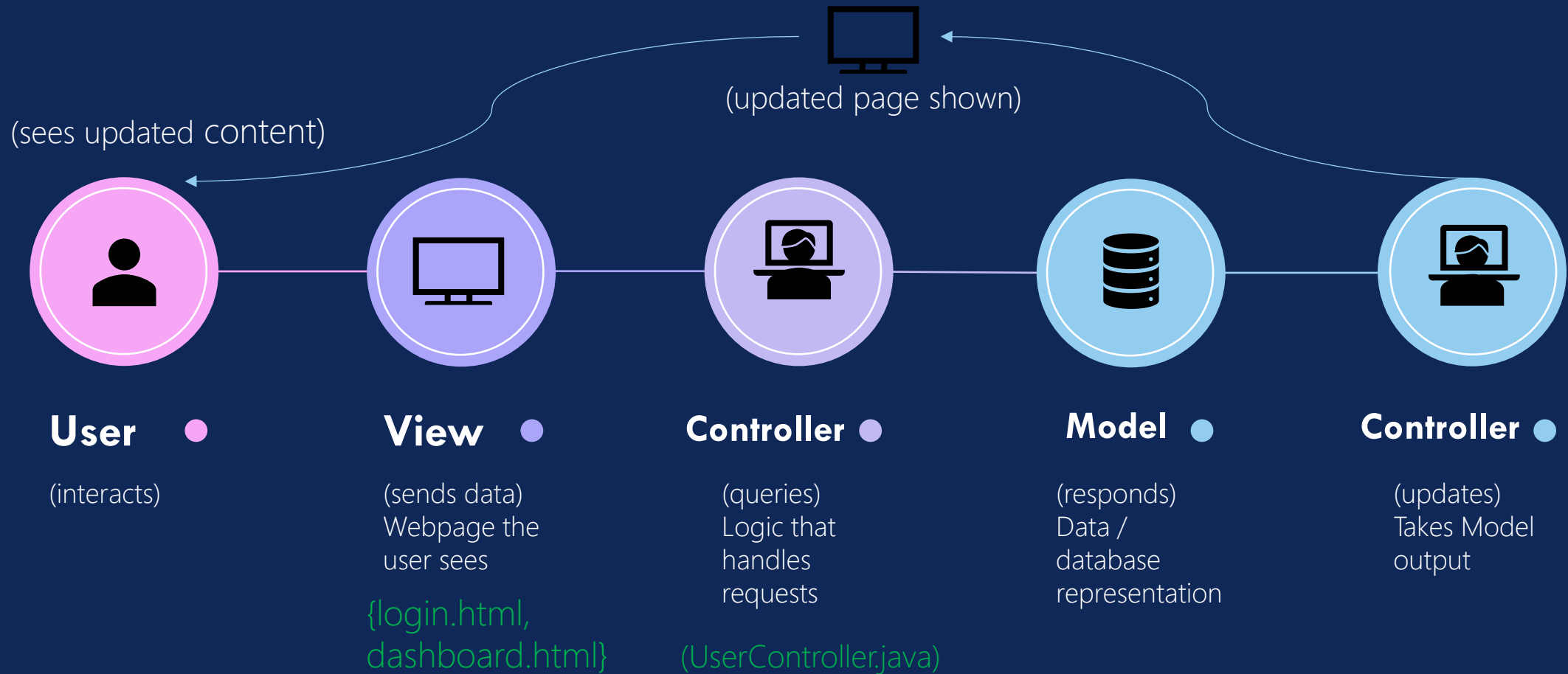
SOFTWARE ARCHITECTURE: MVC PATTERN

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WHAT IS MVC ARCHITECTURE?

- MVC stands for Model-View-Controller.
- It separates an application into three interconnected parts.
 - Model: Manages the data and business logic.
 - View: Displays the data to the user (User Interface).
 - Controller: Handles user input and interactions, updates Model and View.
- This design improves organization, modularity, and scalability.

DETAILED MVC ARCHITECTURE:

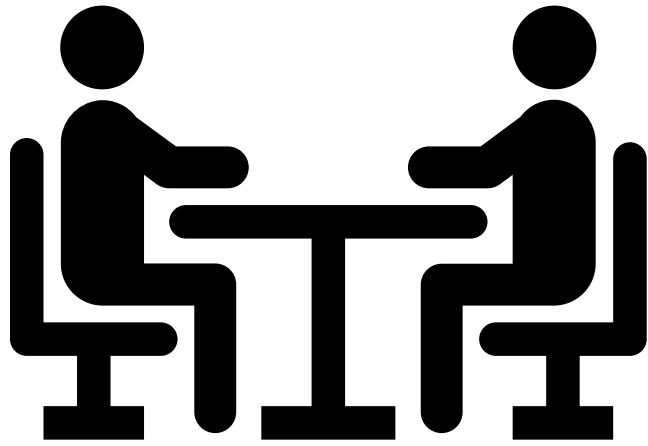


KEY ADVANTAGES OF MVC:

- Clear separation improves maintainability
 - Easier unit testing of components
 - Enables parallel teamwork

SUMMARY: WHY WE CHOSE MVC

- Organized Code Structure
- Easier to Test and Debug
- Faster and Smoother Team Collaboration
- Prepared for Future Expansion



Communication Methods

- Primary methods of communication:
 - Emails – At the start
 - WhatsApp – Text and calls
 - Discord – Calls with screen sharing
- Meeting format:
 - Weekly calls at the start
 - Calls every 2 days (2 weeks prior to deadline)
 - Calls everyday (5 days prior to deadline)

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Strengths, Challenges and Improvements



Strengths:

Had regular meetings to discuss any doubts we had

WhatsApp messages helped us solve any urgent issues we had

Clear segregation of tasks helped prevent overlapping and confusion



Weaknesses:

People not attending some meetings slowed down decision-making

Meetings at the start lacked structure and wasted a bit of time



What to do going forward:

Settings clear agendas before the meeting starts to avoid wasting time

Confirm tasks at the end to avoid miscommunication

Communication Knowledge Gained During the Project



Benefits of Structured Communication:

Settings up clear agendas before the start of each meeting helped maintain focus



Staying Responsive is Essential:

The help of timely replies on WhatsApp helped to keep tasks moving without any delays



Advantages of Clarification and Active Listening:

Clarifying what needed to be done and actively listening to each group member reduced mistakes caused

Allocation of Work

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Allocation of Work: Scrum Masters & Meetings



Rotating Scrum Master during
each sprint



Held regular Scrum Meetings
across the sprint

User Stories: Fair Assignment



Balanced workload
across the group



Matched story
complexity to skills

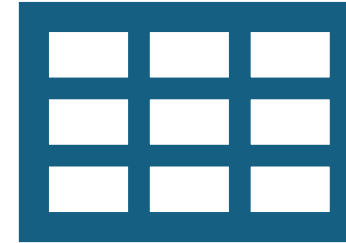


Group review before
assignment

Matching Pace Across the Team



Different paces managed
effectively



Example: Namit assigned multiple
stories during sprint 1

Success Criteria: Did We Meet Them?



All user stories completed



On-time sprint delivery



Effective leadership rotation

Improvements for the Future



More frequent check-ins during
the sprint



Improving task estimation



Realistic plans to take this project onwards

By Denis Gitman

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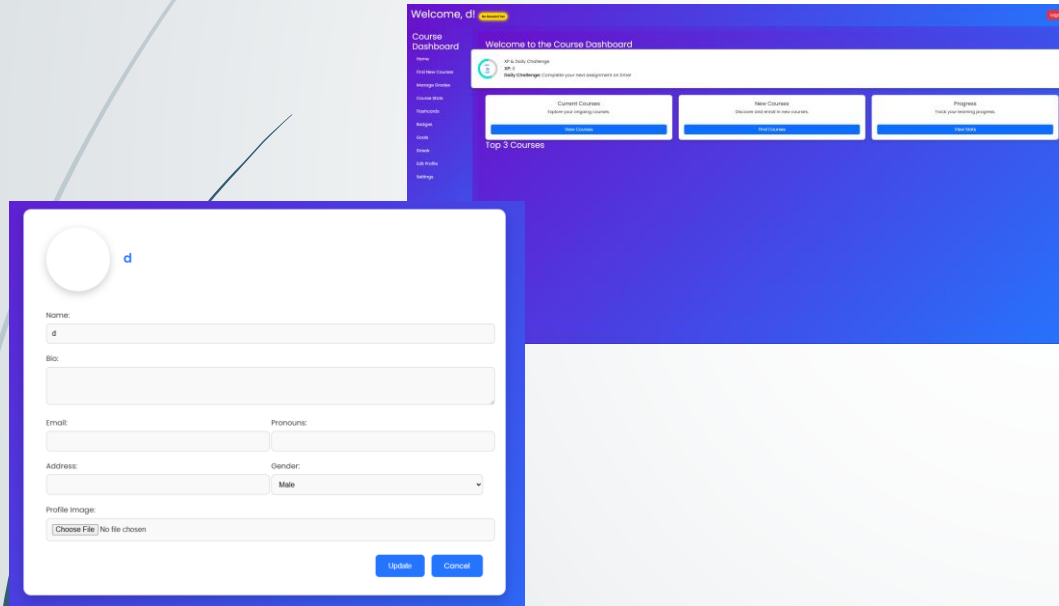
Current Standings & Strengths

User Features:

- Profile Management
- Course Filters
- Personalised Challenges & Goals
- Daily Streak with Mystery Boxes

Design Implementations:

- Dashboard Layout
- Clear Navigation

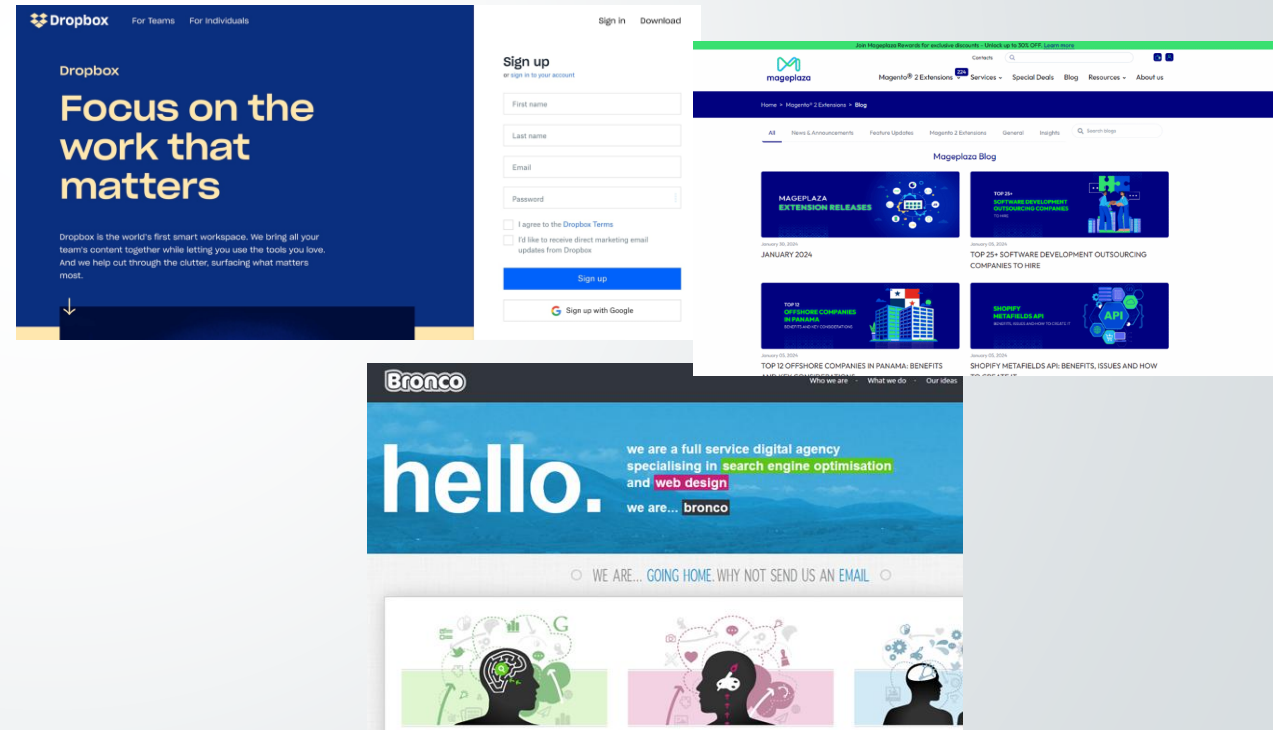


Realistic Next Steps

➤ Development



➤ Design of website layout and user feature implementations



Planned Timeline



Overview of the purpose of the project

By Arshdip Singh

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Overview of the purpose of the project

Encouraging the use of IBM SkillsBuild



Building personal achievements to showcase knowledge and a good record.



Made the platform competitive where friends could rival each other whilst learning.



Provided a gamified structure for courses, making them engaging and unique.



Leaderboard to showcase the standings of a user's network of friends, encouraging them to do more courses.



Overview of the purpose of the project

Helping students to build skills



Create simple UI and navigation system for students to use for a simple access to learning.



Allow students to keep track of their achievements and certificates to boost their overall record.



Guide students through practical, beginner-friendly courses in AI, Blockchain, giving them a strong foundation in real-world tech skills.



Empower students with industry relevant learning paths that align with the most in demand careers in technology and innovation.



Overview of the purpose of the project

Gamification of the web app



Progress tracking to keep a note of a student's learning.



Badges were earned just like achievements in a videogame and streaks could mystery boxes for daily logins.



Students were able to earn XP based on completed courses



Personalised challenges were, to personalize a student's learning journey.



Flashcard game for challenging students in a game based learning.



Social, Legal, and Ethical Issues Raised

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Social Issue: Communication



Problem: Remote
communication issues

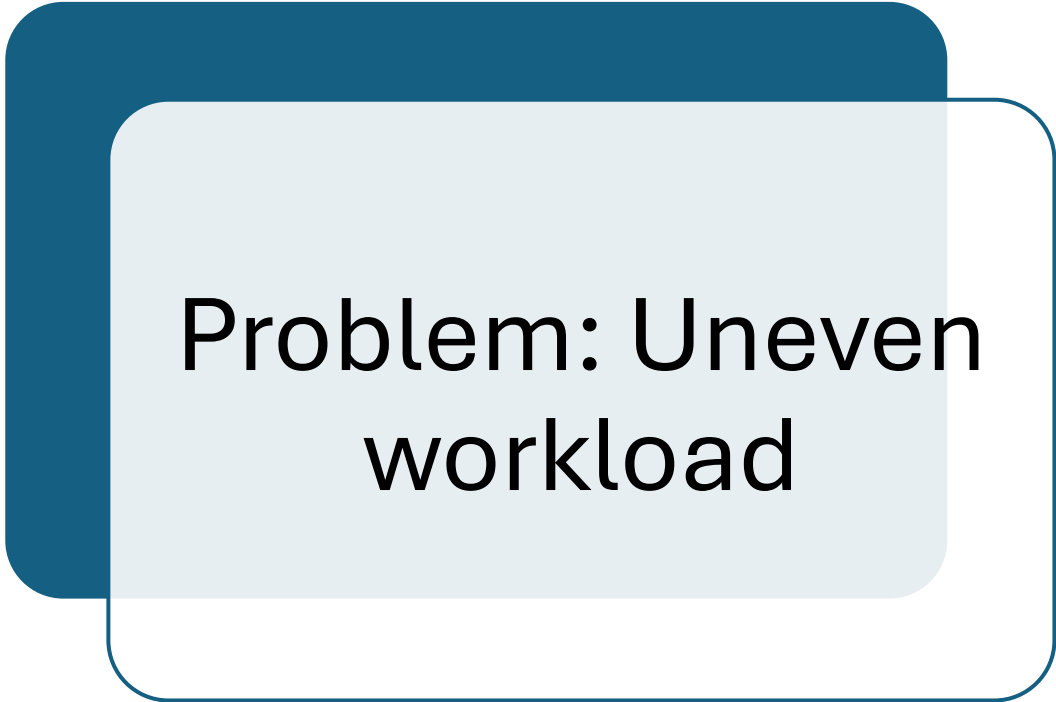


Fix: Use WhatsApp/Discord,
regular check-ins

Social Issue: Accessibility

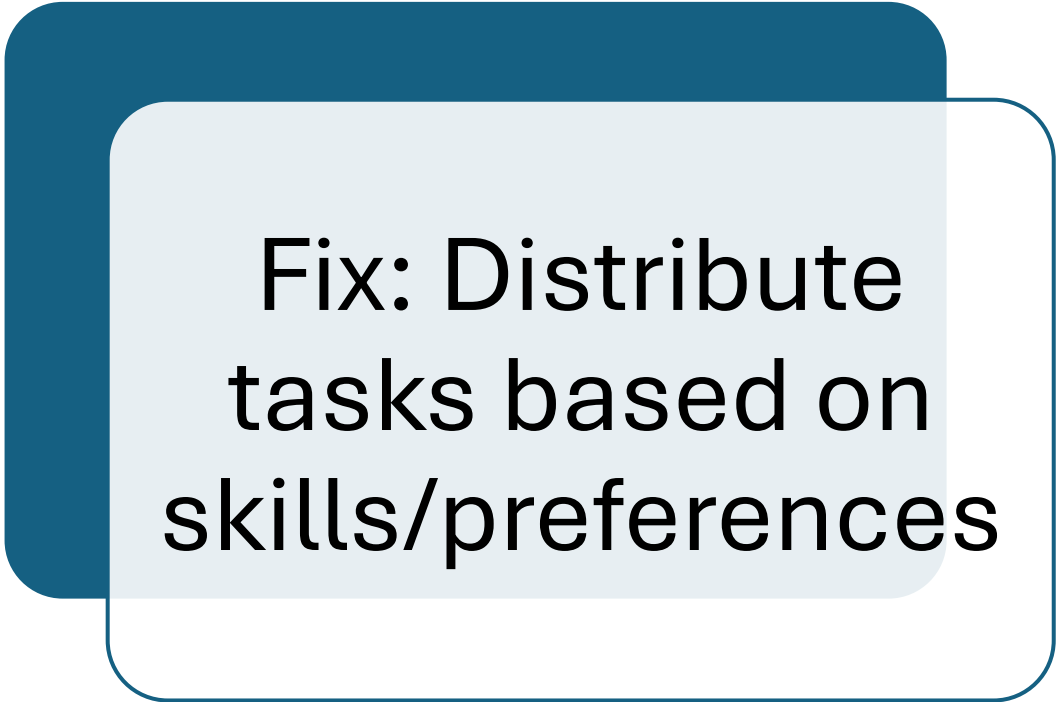
Design must
accommodate
users with
disabilities (e.g.
colour blind)

Social Issue: Workload



Problem: Uneven
workload

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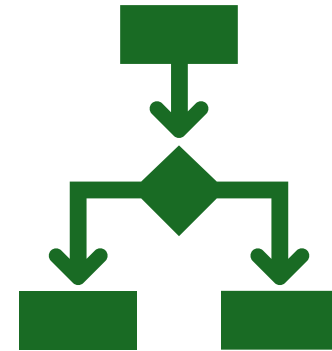
Fix: Distribute
tasks based on
skills/preferences

The diagram consists of a dark blue rounded rectangle with a light blue rounded rectangle inside it. The text 'Fix: Distribute tasks based on skills/preferences' is centered within the light blue rectangle.

Social Issue: Conflict



Problem: Disagreements
amongst team members



Group discussions

Online safety and Respect

Gamification must
promote positive
interaction, and
not unhealthy
competition

Legal: Data protection



PROBLEM: DATA MUST BE
PROTECTED



ENSURE PROPER
ENCRYPTION

Legal: Copyright



PROBLEM: IBM CONTENT
RESTRICTION



FIX: EXTERNAL ASSETS MUST BE
LICENSED OR OPEN SOURCE

Legal: Terms & Accessibility



PROBLEM: LEGAL
COMPLIANCE



NEED TERMS OF
USE/PRIVACY POLICY

Ethical: Over collecting data

Data
minimization

Only collect
information
that's necessary

Ethical: Transparency



TRANSPARENCY WITH USERS



USERS MUST BE CLEARLY
INFORMED HOW DATA IS USED